

Obstructions to Geometric Vertex Decomposability of Toric Ideals of Graphs

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- What is GVD and what are the obstructions to GVD?
- Our theorem and other graphs that are not GVD
- Going forward

What is Geometric Vertex Decomposability?

- **Geometric vertex decomposability (GVD)** is an inductive process that breaks a polynomial ideal down, one variable at a time.
- If a homogeneous ideal is GVD, it has several desirable properties in commutative algebra, including being Cohen-Macaulay and glicci.
- Let $R = \mathbb{K}[x_1, \dots, x_n]$ and fix $y = x_i$ and ideal $I = \langle g_1, \dots, g_m \rangle \subset R$.
- Write $g_i = y^{d_i} q_i + r_i$, where y does not divide any term of q_i .

$$C_{y,I} = \langle q_1, \dots, q_m \rangle \quad N_{y,I} = \langle q_i \mid d_i = 0 \rangle.$$

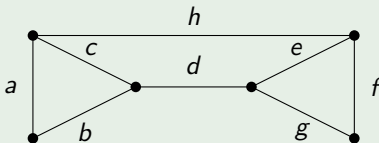
The ideal I is *geometrically vertex decomposable* if I is unmixed, I is squarefree in y (that is, $d_i = 0, 1$ above), and $C_{y,I}$ and $N_{y,I}$ are GVD in $\mathbb{K}[x_1, \dots, \hat{x}_i, \dots, x_n]$.

What is a Toric Ideal of a Graph?

- To every graph G we associate an algebraic object called the **toric ideal of G** , written $I_G \subset \mathbb{K}[E(G)]$.
- I_G is generated by binomials defined by **closed even walks** on G .
- For instance, in the graph below, we get the binomial:

$$(a, b, d, g, f, h) \rightarrow adf - bgh$$

Example



$$I_G = \langle ad^2f - bceg, ce - dh, adf - bgh, bh^2g - acef \rangle$$

GVD Example

Consider a lexicographic ordering such that $c > f > a \dots$

$$I_G = \langle ad^2f - begc, ec - df, adf - bgh, bh^2g - aefc \rangle$$

so that

$$C_{c,I} = \langle beg, e, adf - bgh, aef \rangle = \langle e, adf - bgh \rangle \quad N_{c,I} = \langle adf - bgh \rangle.$$

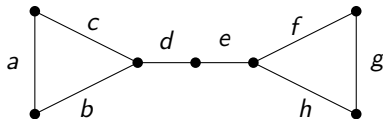
For $J = N_{c,I}$, we can continue to get:

$$C_{f,J} = \langle ad \rangle \quad N_{f,J} = \langle 0 \rangle$$

- One more step for $C_{f,J}$ terminates with the ideal $\langle d \rangle$.
- Similarly for $C_{c,I}$.

When GVD Gets Stuck

- Some ideals get "stuck" partway through the decomposition process — these obstructions are what we will focus on.
- One possible obstruction: N or C ideals are not unmixed (think equidimensional). For example $\langle xy, xz \rangle = \langle x \rangle \cap \langle y, z \rangle$ has mixed dimension components which is a problem.
- The second possible obstruction: persisting **non-squarefree** terms, i.e. terms with higher order variables, like d^2 .
- In the example below, no matter what variable y we choose, C_{y, I_G} is generated by non-squarefree monomial ideal, which is not GVD.



$$I_G = \langle ad^2fh - bce^2g \rangle$$

The Problem

- There is currently no complete combinatorial classification of which graphs produce non-GVD toric ideals.
- Goal: Identify *which graph structures* cause this failure, and why.
- Our approach: Focus on double extended bowtie graphs, prove that critical generators obstruct the GVD property, and test how the obstruction responds to adding edges (using the `GeometricDecomposability` package in Macaulay2 for example).

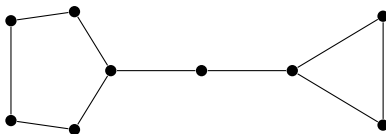
Theorem (Cummings, Da Silva, Rajchgot, Van Tuyl, 2023)

- 1 *The toric ideal I_G is GVD if G is bipartite.*
- 2 *I_G being GVD is preserved under gluing an even cycle along one edge.*
- 3 *If I_G has a squarefree degeneration, then I_G is conjectured to be GVD.*

Critical Bowtie Generator

Definition

A **critical bowtie generator** corresponding to an k -extended bowtie graph (with $k \geq 2$) is a binomial $\Gamma = a^2u - b^2v$ appearing in the reduced Gröbner basis of I_G with respect to a lexicographic order $<$, where u, v are monomials not involving a or b .



Proposition

If a term of a primitive binomial γ of G divides a term of $\Gamma = a^2u - b^2v$, then γ must involve edges in the induced subgraph of Γ .

Issue with General Non-squarefree Degeneration

What could go wrong with using $\Gamma = a^2u - b^2v$ to show that an ideal is not GVD?

Example

Suppose $I = \langle u^2ab - v^2cd, ub - st, bw^2 - afg, \dots \rangle$ with the order $a > s > \dots$ yielding that $u^2ab - v^2cd$ as a necessary generator.

$$J = C_{a,I} = \langle bu^2, ub - st, fg, \dots \rangle$$

At this stage, we can change the order to $b > \dots$ and get

$$C_{b,J} = \langle u^2, u, fg, \dots \rangle$$

And the $u^2ab - v^2cd$ generator no longer persists in the remaining ideals.

Theorem: Double-extended Bowties Obstructions

Theorem

Suppose that G contains an induced subgraph which is isomorphic to a k -extended bowtie graph with $k \geq 2$. Then I_G is not geometrically vertex decomposable.

Proof Sketch

- If there exists an induced extended bowtie subgraph, then Γ is part of a universal Gröbner basis of I_G .
- Try to define a geometric vertex decomposition using y .
- If y is not in the support of Γ , then Γ will persist in the N -ideal.
- If y is in the support of Γ , then a non-squarefree term will persist in the C -ideal. Cancellation can't occur since this would contradict the universality assumption on Γ .

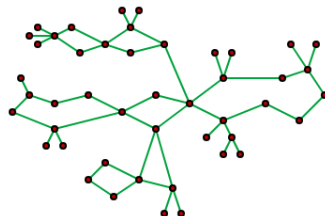
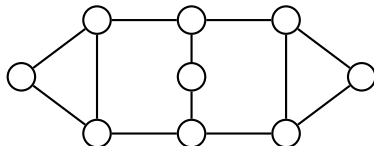
Families of Graphs Which are not GVD

Theorem

Let G be one of the following graphs:

- 1 Odd asymmetric Barbell graphs (two odd complete graphs K_n with $n > 3$ joined by an edge)
- 2 Certain Cactus graphs
- 3 Ha-Kara-O'Keefe graphs

Then I_G is not GVD.



Question: Can we add an edge to break the obstruction? That is, what if G contains an extended bowtie as a subgraph, but not as an induced subgraph?

- This will be discussed in Munso's talk.

Thank you!